Annotation Guide: Aspect-based sentiment analysis (sentence-based)

1. Introduction

Aspect-based sentiment analysis is a method for analyzing and evaluating opinions and feelings in text that focuses on specific aspects or features of a product, service or topic. These guidelines define how annotators should analyze texts to extract sentiment-related information focused on these specific aspects. The purpose of this annotation is to recognize aspects and their sentiment polarity in sentences from reviews of software products. The annotator's task is to identify the following sentiment elements:

- **Aspect Category**
  Only aspects or aspect terms that can be assigned to the following aspect categories or are related to them should be considered in the sentences:

Ease of Use	Opinions related to the usability and user-friendliness (such as handling, convenience and flexibility) of the application/software.
Technical Performance	Opinions related to the technical implementation (such as performance, speed, errors and actuality) of the application/software.
Interface Design	Opinions that relate to the interface / design (such as structure, appearance and overview) of the application / software.
Functional Scope	Opinions related to the range of functions, specific functions, program support and the functionality of the application/software. The functionality must go beyond the basic functions.
Customer Support	Opinions related to communication with customer support, the availability of customer support, or the effectiveness of customer support.
Pricing	Opinions related to the prices or the price-performance ratio of the application/software.
General Experience	Opinions about the application / software in general.

- **Aspect Term / Phrase**
  Single or multi-word terms (aspect phrase) that represent the target of the expressed sentiment.

  Examples:

*Poor **accessibility** - long **waiting times***

"Accessibility" and "waiting times" are aspect phrases and can be assigned to the "Customer support" aspect category.

Aspect Term: **"accessibility "**
Aspect Category: **Customer Support**
Aspect Polarity: **Negative**

Aspect Term: **"waiting times"**
Aspect Category: **Customer Support**
Aspect Polarity: **Negative**

Important:
It is also possible for an aspect category to be addressed implicitly in a sentence, which means that there is no aspect phrase for an aspect category in the sentence. The following examples illustrate this:

Instead of becoming clearer, everything becomes more confusing

Here, the aspect category "Ease of Use" expresses a negative sentiment, but without a specific aspect phrase as the target of the sentiment.

Aspect Term: **"NULL" (implicit)**
Aspect Category: **Ease of Use**
Aspect Polarity: **Negative**

*Fast **feedback**, efficient to process, clearly laid out*

In this example, a positive sentiment is expressed towards the phrase "feedback" ("Customer Support" category). Furthermore, Ease of Use and the Interface Design are rated positively, but without reference to a specific phrase.

Aspect Term: **"feedback"**
Aspect Category: **Customer Support**
Aspect Polarity: **Positive**

Aspect Term: **"NULL" (implicit)**
Aspect Category: **Ease of Use**
Aspect Polarity: **Positive**

Aspect Term: **"NULL" (implicit)**
Aspect Category: **Interface Design**
Aspect Polarity: **Positive**

- **Aspect Polarity**
  Each identified aspect (or aspect term) must be assigned one of the following polarities P = {**Positive, Negative, Neutral**} based on the mood expressed towards it in the sentence:

  - **Positive and negative**: An aspect term should be categorized as Positive or Negative if the sentence expresses a Positive, Negative or both Positive and Negative attitude, opinion, evaluation, emotion or feeling etc. of an opinion holder towards the aspect term/aspect.

  - **Neutral:** An aspect term should be classified as neutral if a neutral sentiment is expressed with regard to the aspect term.

    i. Good connection of the user interfaces -> {Positive}
    ii. Login was not possible -> {Negative}
    iii. {function name} is OK.-> {Neutral}

Example:

*The **assistance** is often not clear; everything is complicated*

1. Aspect in the example:

The clarity of the instructions is rated negatively and therefore relates to the Ease of Use aspect of the application. The aspect term or the goal of the sentiment is explicitly referenced in the text.

Aspect Term: **"assistance"**
Aspect Category: **Ease of Use**
Aspect Polarity: **Negative**

2. Aspect in the example:

The complexity of the application is rated negatively and therefore refers to the Ease of Use aspect of the application. However, no specific aspect term is referred to.

Aspect Term: **"NULL" (implicit)**
Aspect Category: **Ease of Use**
Aspect Polarity: **Negative**

2. Annotation Guidelines: Aspect Terms

a) What is an aspect term?

- Nouns that are explicitly evaluated with sentiment in the sentence. Note the 1st example, where there is only one **aspect term** ("{function name}"), which is evaluated as a coherent term.

*You can work well with **{function name}***

Aspect Term: **"{function name}"**
Aspect Category: **Functional Scope**
Aspect Polarity: **Positive**

*great **service***

Aspect Term: **"service"**
Aspect Category: **Customer Support**
Aspect Polarity: **Positive**

- An aspect term should only be marked if the author expresses a sentiment towards it.

Examples:

I have been working with this program for several years

Here, program is not marked as an aspect term, as no sentiment is expressed towards the phrase.

*I have been working with this **program** for several years and I still like it very much.*

Aspect Term: **"program"**
Aspect Category: **General Experience**
Aspect Polarity: **Positive**
Reason: In the second example, a positive sentiment is expressed towards the aspect term, which is why the term is marked as an aspect phrase.

- **Phrases should be annotated as detailed as possible.** Any additional information that further specifies the aspect should be included in the phrase.

Examples:

There are a lot of temporary failures in the software;

Aspect Term: ***"failures in the software"***
Aspect Category: **Technical Performance**
Aspect Polarity: **Positive**

*[...], **Networking with {function name}** is good, [...]*

Aspect Term: **"Networking with {function name}"**
Aspect Category: **Functional Scope**
Aspect Polarity: **Positive**

*The **automatic {function name}** could be better implemented*

Aspect Term: **"*automatic {function name}*"**
Aspect Category: **Functional Scope**
Aspect Polarity: **Negative**

*If the **{program name} tool** crashes, a new file has often had to be created.*

Aspect Term: **"*{program name} tool*"**
Aspect Category: **Technical Performance**
Aspect Polarity: **Negative**

***Functions in {function name}** are much better implemented, especially compared to before*

Aspect Term: **"Functions in {function name} "**
Aspect Category: **Functional Scope**
Aspect Polarity: **Positive**

b) Temporal information in the phrase is not annotated:

*If the **{function name}** is used for more than 6 hours, it is no longer updated automatically and errors occur.*

Aspect Term: **"*{function name}*"**
Aspect Category: **Functional Scope**
Aspect Polarity: **Negative**

c) Fixed phrases are not split:

***Copy & Paste** works very well!*

Aspect Term: **"*Copy & Paste*"**
Aspect Category: **Functional Scope**
Aspect Polarity: **Positive**

d) Numbers that belong to an aspect term are included in the phrase.

*The **font size 32** for headings is too large.*

Aspect Term: **"font size 32"**
Aspect Category: **Interface Design**
Aspect Polarity: **Negative**

e) Multiple sentiment statements that refer to the same aspect term.

Aspect terms that are provided with several sentiment specifications and refer to the same aspect are only annotated once. If different categories are addressed by the sentiment information, the aspect term is annotated multiple times.

*Very fast and legible **processing***

Aspect Term: **"processing"**
Aspect Category: **Ease of Use**
Aspect Polarity: **Positive**

*The **{function name}** is well designed and easy to use!*

Aspect Term: **"{function name}"**
Aspect Category: **Ease of Use**
Aspect Polarity: **Positive**

Aspect Term: **"{function name}"**
Aspect Category: **Interface Design**
Aspect Polarity: **Positive**

f) Indicators of subjectivity (i.e. words/phrases that express opinions, evaluations, etc., such as "beautiful", 'great', or "clear") are NOT considered aspect terms or components of aspect terms.

*very exemplary **support***

Aspect Term: **"support"**
Aspect Category: **Customer Support**
Aspect Polarity: **Positive**

*Good and easy to **use**, great **service** in case of uncertainties or errors.*

Aspect Term: **"use"**
Aspect Category: **Ease of Use**
Aspect Polarity: **Positive**

Aspect Term: **"service in case of uncertainties or errors" (long)**
Aspect Term: **"service" (short)**
Aspect Category: **Service**
Aspect Polarity: **Positive**

Adjectives bordering the phrase are generally not marked.

g) **The identified aspect terms should be annotated as they are, even if they are misspelled.**

*I am satisfied with the **aplication**, but not with the **suport**.*

Aspect Term: **"aplication"**
Aspect Category: **General Experience**
Aspect Polarity: **Positive**

Aspect Term: **"suport"**
Aspect Category: **Customer Support**
Aspect Polarity: **Positive**

h) **Pronouns (e.g. "it", "they", "this") should not be marked as aspect terms even if they refer to an aspect. For example, "it" is not annotated in the following examples. Here, the aspect should be annotated as "implicit".**

*The error was quickly rectified and it was a pleasant, almost funny, **conversation**.*

Aspect Term: **"conversation"**
Aspect Category: **Customer Support**
Aspect Polarity: **Positive**

Unfortunately, it didn't get any better.

Aspect Term: **"NULL" (implicit)**
Aspect Category: **General Experience**
Aspect Polarity: **Negative**

i) **Terms that generally address one of the six aspect categories, such as "Customer Support" or "Functional Scope" are also regarded as aspect terms.**

*That's great - the **customer support** is outstanding!*

Aspect Term: **"customer support"**
Aspect Category: **Customer Support**
Aspect Polarity: **Positive**

***Pricing** has been very good for years.*

Aspect Term: **"Pricing"**
Aspect Category: **Pricing**
Aspect Polarity: **Positive**

j) **If an aspect term occurs more than once in a sentence, only its first occurrence in the sentence should be annotated.**

Customer support *is not always available - although the times have improved - but customer support is still a great help.*

Aspect Term: **"Customer support"**
Aspect Category: **Customer Support**
Aspect Polarity: **Negative**

k) Dealing with implicit aspects:

Implied aspects are aspects that are not explicitly mentioned but can be derived from adjectives or other expressions.

Generally very positive; unfortunately a few new ***functions*** *are not immediately apparent.*

Aspect Term: **"NULL" (implicit)**
Aspect Category: **General Experience**
Aspect Polarity: **Positive**
Reason: In this example, "Generally" can be understood as an implicit reference to the aspect category "General Experience".

Aspect Term: **"functions"**
Aspect Category: **Ease of Use**
Aspect Polarity: **Negative**
Reason: In addition, the aspect category "Ease of Use" is referenced with the phrase "functions".

Multiple statements in relation to the same aspect category and with the same sentiment should only be marked with one annotation.

Visually pleasant and easy to understand

Aspect Term: **"NULL" (implicit)**
Aspect Category: **Interface Design**
Aspect Polarity: **Positive**

Reason: The aspect category "Interface Design" is implied here and evaluated as "Visually pleasant" and "Understandable". A unique annotation is made, as both refer to the same implied aspect phrase.

Important: Implicit aspects should also be annotated.

l) Out of Scope:

- Statements that only refer to competitors and are not valid on their own are not annotated.
- Feedback relating to the completion of the feedback form is not annotated
- Reviews that are not in German are not annotated.

m) Annotation of multiple aspect terms:

- **Aspect specificatons with examples:**

  The superordinate terms and all examples listed should be annotated.

*Some of the **functions** such as **{function name}** are no longer available since the latest {program name} update*

  Aspect Term: **"functions"**
  Aspect Category: **Functional Scope**
  Aspect Polarity: **Negative**

  Aspect Term: **"{function name}"**
  Aspect Category: **Functional Scope**
  Aspect Polarity: **Negative**

- **Annotation of terms that are separated by an "and":**

  Both terms will be annotated individually here.

*[...], good integrated **functions** and **analyses***

  Aspect Term: **"functions"**
  Aspect Category: **Functional Scope**
  Aspect Polarity: **Positive**

  Aspect Term: **"analyses"**
  Aspect Category: **Functional Scope**
  Aspect Polarity: **Positive**

- **Annotation of terms that are separated by a "/":**

  The entire term should be annotated here if there is no space between the first term and the separator.

*The view in **{function name}/{alternative function name}** is a bit vague.*

  Aspect Term: **"{function name}/{alternative function name}"**
  Aspect Category: **Functional Scope**
  Aspect Polarity: **Negative**

n) Annotation of related phrases that are linked by "+":

*Great **overview + design + interface**.*

Aspect Term: **"overview + design + interface"**
Aspect Category: **Interface Design**

Aspect Polarity: **Positive**
Reason: Several aspects of the interface are addressed here.

3. Additional Examples

In each sentence, all aspect terms contained therein, their superordinate aspect categories and the mood (polarity) expressed in relation to them should be annotated, provided they fulfill the conditions listed in chapter 2.

Ideal and really understandable

Aspect Term: **"NULL" (implicit)**
Aspect Category: **Ease of Use**
Aspect Polarity: **Positive**

very understandable.

Aspect Term: **"NULL" (implicit)**
Aspect Category: **Interface Design**
Aspect Polarity: **Positive**

*All tasks can be processed by **{function name}** without any problems.*

Aspect Term: **"{function name}"**
Aspect Category: **Functional Scope**
Aspect Polarity: **Positive**

***Offers** are well structured, informative and easy to access.*

Aspect Term: **"Offers"**
Aspect Category: **Ease of Use**
Aspect Polarity: **Positive**
Reason: Aspect Term is described as understandable and easy to access.

Aspect Term: **"Offers"**
Aspect Category: **Interface Design**
Aspect Polarity: **Positive**
Reason: Aspect Term is described as well structured.

*{**program name**} is clear and easy to use.*

Aspect Term: **"{program name}"**
Aspect Category: **Ease of Use**
Aspect Polarity: **Positive**
Reason: Only the operation of the application is described, but not the interface design (although the key word "clear" is used).

*I would like to see better **automation** in the application.*

Aspect Term: **"automation"**
Aspect Category: **Ease of Use**
Aspect Polarity: **Negative**
Reason: The lack of automation in the application hinders usability/technical performance during use.

It's too technical and complicated/you often have to rethink things.

Aspect Term: **"NULL" (implicit)**
Aspect Category: **Ease of Use**
Aspect Polarity: **Negative**
Reason: The application is described as too complicated and not intuitive enough. Equal sentiments towards the same implicitly addressed aspect category are summarized in an annotation.

4. Annotations tool:

The sentence to be annotated is displayed at the top. First, a label can be selected (e.g. as here Technical Performance and Negative) and used to mark a point in the text. Hotkeys can also be used in the program for annotation. These are available in the lower part of the application under "Settings -> Hotkeys".

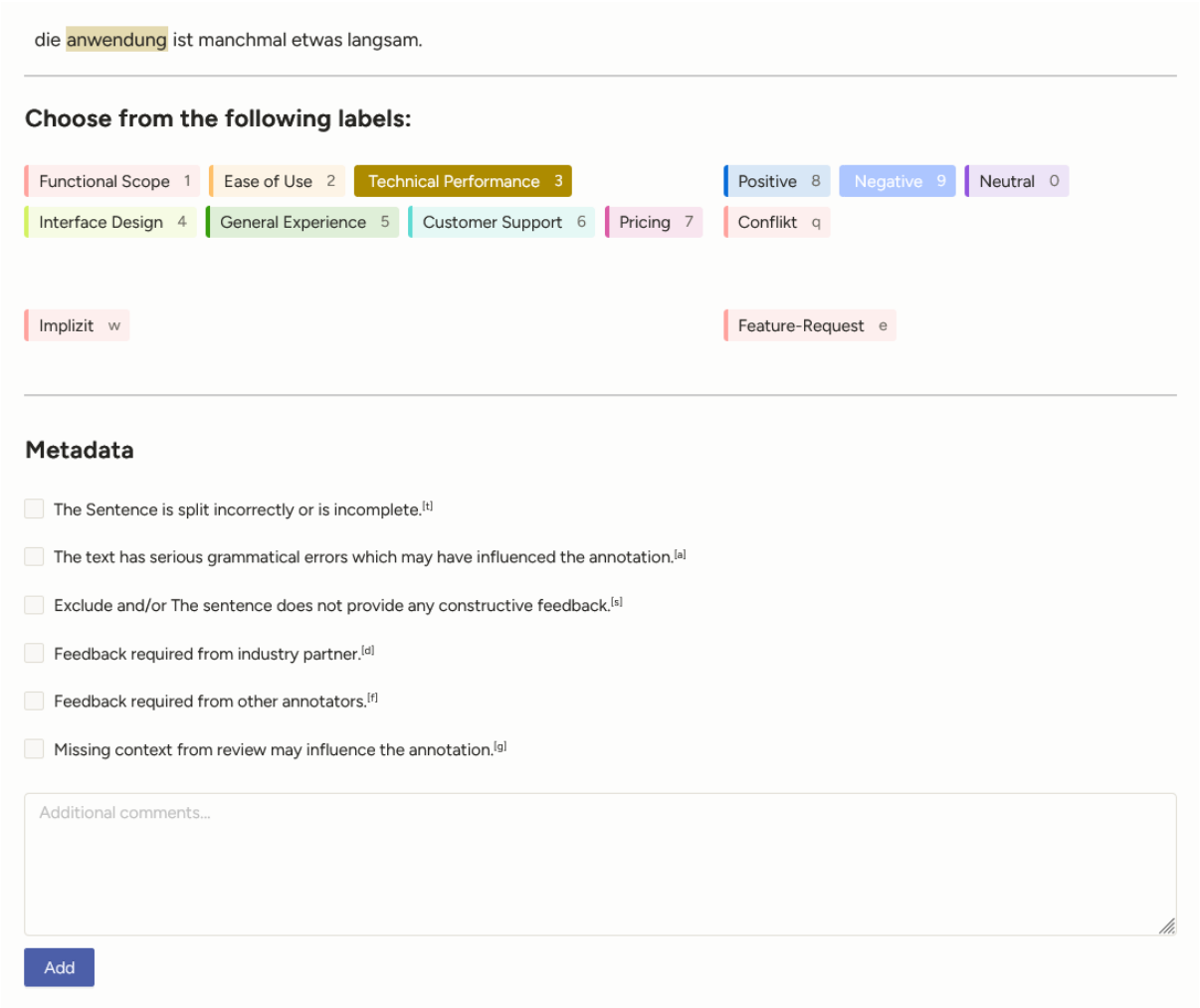

Figure 1: Annotation interface in the Label Studio program

If it is an implicit aspect, the corresponding label (aspect category + aspect sentiment) can be selected together with the "implicit" label and any area in the text can be marked (here in the example price-performance, negative + implicit). The marked text is replaced by a placeholder after the annotation.

The following metadata can also be specified:

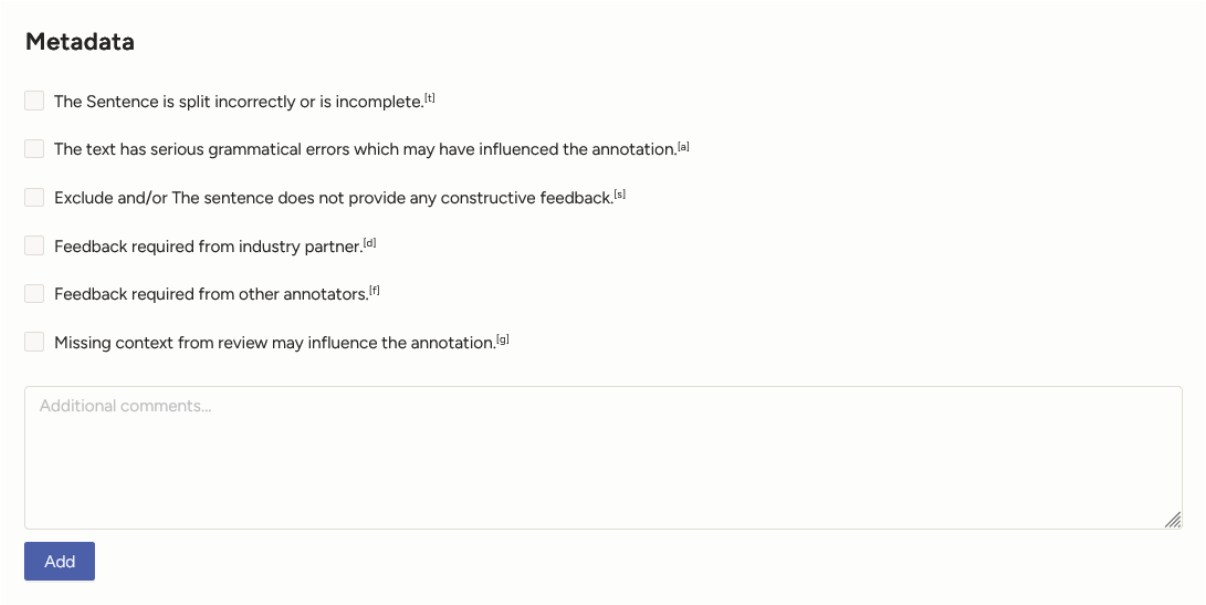

Figure 2: Additional metadata for annotation in Label Studio

1. Checkbox Metadata (Split):
If the review was incorrectly split into individual sentences or the sentence is not complete.

2. Checkbox Metadata (Spelling):
If the review contains serious grammatical errors or spelling mistakes, so that the annotation can only be carried out with reservations.

3. Checkbox Metadata (Unconstructive criticism):
If, in the opinion of the annotator, only unconstructive criticism, insults or pointless statements are expressed in the review. The review should not be included in the data set.

4. & 5. Checkbox Metadata (Feedback annotators/industry partners):
If the annotators are not yet sure about the annotation, a note can be left for a further review. In the first case, feedback from the industry partner is desired; in the second case, consultation with the other annotators should take place.

6. Checkbox Metadata (Missing information):
If important information is missing in the review in order to make a meaningful annotation, e.g. Reference is made to something that does not occur in the current sentence. Information is missing to be able to categorize the statement.

7. Comment field:
Here you can leave additional information for other annotators if there are discrepancies or questions about the review.

Once all aspects have been annotated, you can click on *Submit* (blue button, bottom right) to continue with the annotation of the next feedback.